

CHEMISTRY PAPER 26 | UNIT 11 PART 1

Environmental Chemistry: Air (sections 11.1 and 11.2)

Syllabus: Unit 11 Environmental Chemistry Air, sections 11.1 and 11.2.1 to 11.2.6, her book p153 to 156 (27 marks) + **Revision:** Unit 8 Energetics, enthalpy of reaction and bond energy, her book p131 to 132 (11 marks)

Total Marks: 38

Time Allowed: 1 hour 30 minutes

Version A

Note (Seemab ke liye): Is parchay ka bara hissa (27 number) **Unit 11 ka Part 1** hai, wohi jo kal aap ke notes aur audio mein tha. **Notes ke teeno Quiz sawal Section B mein hain.** Baqi 11 number revision ke hain: **Unit 8, bond energy.** Wo isi liye hai ke Chemistry exam-25 mein aap ka sirf ek bara nuqsan hua tha, Q.7 mein, jahan reaction ulti thi aur ΔH ka ishara badalna tha.

HIDAYAAT (Instructions)

1. Saaray sawal hal karne hain. Koi optional sawal nahin hai.
2. Section A mein sirf apne option ka letter likhein, jaise 1. (b). Aakhir mein gin lein ke dason ho gaye hain, **kisi ko khaali mat chhorein.**
3. Har calculation mein **pehle formula, phir values, phir jawab.** Marks working ke hain.
4. **Har jawab ke saath uska unit likhein.** Unit na likhne par aadha mark katega. Exam-25 mein aap ka aadha mark isi wajah se gaya tha.
5. Jahan **definition** maangi gayi ho wahan kitab ke apne alfaaz likhein.
6. Calculator use kar sakti hain.

SECTION A: MULTIPLE CHOICE QUESTIONS (10 × 1 = 10 MARKS)

1. The envelope of gases and water vapour surrounding the planet Earth is called the:

(a) troposphere	(b) ozone layer	(c) atmosphere	(d) biosphere
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2. According to your book, the percentage of nitrogen in dry air is approximately:

(a) 78%	(b) 21%	(c) 4%	(d) 1%
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3. In a tropical rainforest the air may contain water vapour up to:

(a) 1%	(b) 10%	(c) 21%	(d) 4%
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4. Which of these is a **secondary** pollutant, that is, one not emitted directly?

(a) sulphur dioxide	(b) ground level ozone	(c) methane	(d) carbon monoxide
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5. Carbon monoxide combines with haemoglobin in the blood to form:

(a) oxyhaemoglobin	(b) methaemoglobin	(c) carboxyhaemoglobin	(d) carbonic acid
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6. Carbon monoxide is mainly produced when:

(a) there is not enough oxygen for complete combustion	(b) there is excess oxygen	(c) sunlight acts on nitrogen oxides	(d) vegetation decomposes
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7. The two oxides of nitrogen named in your book, collectively written as NO_x , are:

(a) N_2O and NO	(b) NO_2 and N_2O_5	(c) N_2O and NO_2	(d) NO and NO_2
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8. [Revision, Unit 8] The sign of the enthalpy of reaction for an **exothermic** reaction is:

(a) always zero	(b) negative	(c) positive	(d) sometimes positive, sometimes negative
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9. [Revision, Unit 8] Standard enthalpy change is measured at:

(a) 0°C and 2 atm	(b) 100°C and 1 atm	(c) 25°C (298 K) and one atmospheric pressure	(d) 25 K and 1 atm
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10. [Revision, Unit 8] According to your book, the higher the activation energy of a reaction, the:

(a) lower the reaction rate	(b) higher the reaction rate	(c) more exothermic the reaction	(d) larger the enthalpy change
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SECTION B: SHORT ANSWER QUESTIONS (3 × 4 = 12 MARKS)

Teenon sawal hal karein. Ye teenon aap ke Unit 11 Part 1 notes ke Quiz sawal hain.

2. Define **atmosphere** and **pollutant** in the words of your book. (2) Then state the percentage of nitrogen and of oxygen in dry air, and the maximum percentage of water vapour your book gives for a tropical rainforest. (2) 4
3. Name the **six** air pollutants shown in Figure 11.2. ($1\frac{1}{2}$) For **sulphur oxides** and for **nitrogen oxides**, write their chemical formulas, one source of each, and two harmful effects of each. ($2\frac{1}{2}$) 4
4. (i) How is carbon monoxide produced, and why can a person not detect its presence? ($1\frac{1}{2}$) (ii) Explain what it does inside the body, naming the compound it forms with haemoglobin. ($1\frac{1}{2}$) (iii) Why does your book call ground level ozone a **secondary** pollutant? (1) 4

Dono sawal hal karein. Har sawal 8 marks ka hai. Q.5 Unit 11 se hai aur Q.6 revision (Unit 8) se.

5. Air pollutants.

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(i) Copy and complete this table for **any four** of the six air pollutants in your book. (4)

Pollutant	One source	One harmful effect

(ii) Explain what **incomplete combustion** means, and state which pollutant it produces. (2)

(iii) A gas heater is left burning in a closed room with the windows shut. Using your answer to (ii), explain why the room becomes **more** dangerous the longer the heater burns, and state one simple precaution. (2)

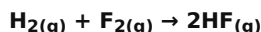
6. [Revision, Unit 8: Enthalpy and bond energy]

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(i) Define **enthalpy of reaction** in the words of your book, and state its sign for an exothermic and for an endothermic reaction. (2)

(ii) Write your book's formula for calculating ΔH° from bond energy data. (1)

(iii) Hydrogen reacts with fluorine to form hydrogen fluoride:



The bond energies are: H-H = 436 kJ/mol, F-F = 155 kJ/mol, H-F = 567 kJ/mol.

Calculate ΔH° for this reaction, showing the formula, the substitution and the answer with its unit. (3)

(iv) State whether this reaction is exothermic or endothermic, and give the reason. (1)

(v) Now consider the **reverse** reaction: $2\text{HF}(\text{g}) \rightarrow \text{H}_2(\text{g}) + \text{F}_2(\text{g})$. Write its ΔH° with its sign and unit, and explain in one sentence why it is what it is. (1)

END OF QUESTION PAPER

Units covered: Unit 11 Environmental Chemistry Air, sections 11.1 and 11.2 (MCQs 1 to 7, Questions 2, 3, 4 and 5) | Revision: Unit 8 Energetics (MCQs 8, 9, 10 and Question 6).

Student's Name: _____ Date: _____ Marks Obtained: _____ / 38